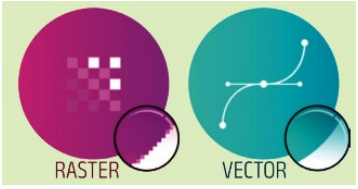
# Chapter 1 - Design Factory

1. Two types of images: Vector & Raster.

**Vector images** are created using mathematical equations that define the lines and shapes. These are easy to edit, to scale without loss of quality, and with small file size.
SVG and EPS are vector file formats.
[Inkscape and LibreOffice Draw are vector image editing software].

Since **Raster images** are made up of pixels, they may become pixelated or lose quality when scaled.
JPEG and PNG are examples of Raster file formats.
[GIMP and Krita are Raster image editing tools].




2. Free software for graphic editing tasks:
GIMP, Krita, Inkscape, Scribus, and LibreOffice Draw.

3. **Inkscape** is widely used to design logos, illustrations, posters, brochures, and typography with accuracy and creativity.
Inkscape's default file format is Scalable Vector Graphics (SVG).
Inkscape supports file formats such as SVG, AI, EPS, PDF, PS, and PNG.

4. To prepare the canvas in Inkscape:
**File** menu → **Document Properties** window (shortcut: Shift+Ctrl+D)
5. To make background for the poster: **Create Rectangle & Squares** (R)
6. To position this square exactly on the page, set the **X, Y values to 0** (zero) in the formatting toolbar.
7. To select objects and change its size: **Select and Transform Objects** (S).
8. To change the Stroke colour, select the object and hold down the Shift key while selecting a colour.
9. To add the text: **Create & edit text objects** (T)
10. To draw circle, ellipse or arc: **Create circles, ellipses, and arcs** (E):
11. To create regular lines, curved shapes, logos, illustrations, and fonts: **Bezier Curve Tool** (Pen tool) (B)
12. To curve the line on both sides: **Edit path by nodes** (N).
13. To copy or get a duplicate Edit → Duplicate (Ctrl+D).
14. To remove the colour assigned to objects:- use the X at the left end of the colour palette.
15. Object – Fill and Stroke : Colours can be applied to both the interior and the border of shapes.
16. The **Stroke style** window allows choosing different types of borders and adjusting their thickness. The **Blur** option in the same window can be used to blend edges and bring shadows and a 3D effect to designs. The **Opacity** option is used to control the transparency of colours, making shapes partially or fully visible.
17. Path – Intersection: To retain the common part of two objects (eg:- circles).
18. Object-Group (Ctrl+G): To group the selected objects to one object.
19. Pg Up and Pg Dn keys:- to bring the selected object to the top or bottom.
20. Object - Align and Distribute:- to align and arrange multiple shapes as desired.
21. To save an inkscape file to PNG image:- File → Export PNG Image.

## Chapter 2 - Let us prepare a Newspaper

1. Desktop publishing (DTP) software:
   Scribus, LaTeX, Adobe InDesign, Microsoft Publisher and QuarkXPress.
2. **Scribus** : A free, open-source desktop publishing software, with multi-language support.
   (Main tools include the Shape tool, Text Frames for content and Image tool for image)
3. To start a new page in Scribus: **NewDocument → Single page**
4. To draw a square on the canvas : use the **Shape** tool
5. To modify the shape: use **Node Editing**
6. To take a duplicate of shape: **Item → Duplicate/Transform → Duplicate**.
7. To add text: use the **Text Frame** tool
   (Handle Text Overflow when text is too long)
8. To draw a straight line: use the **Line** tool and press **Ctrl** key while drawing.
9. To include images: use the **Image frame** tool and drag to get a frame → Right click on the frame, **Content → Get Image**
10. A QR Code is a two-dimensional barcode.
    To create a QR Code of the page: Click on the **Barcode** icon → Make apt changes.
11. To arrange texts in columns: **Text Frame** to drag a frame → **Columns & Text Distance**
12. To arrange images in between the texts:
    Select the image → **Shape** → **Text flow around frame**
    (If the image is below the texts, **Level → Raise to Top**)
13. To arrange letters in various ways, use the **Text Flow**
14. To export as PDF, **File → Export**.



## Chapter 3 – Computer Language

1. To represent **ON** and **OFF** states of computer, we use the binary, **1** and **0**.
   (computers can detect ON (1) and OFF (0) signals only)
2. Compilers and interpreters are **translators** used for converting human language into binary.
   Compilers convert the entire program into binary code.
   Interpreters translate and execute code line by line, requiring the source code each time.
   (Python uses interpreter as its translator)
3. Computer programming languages: Python, Java, C, C++, Ruby, PHP ...
4. **Python** is an open-source, high-level language with simple syntax.
   (Developed by Guido van Rossum in 1990).
5. Algorithm?
   A sequenced set of step-by-step instructions, to solve a problem logically.
6. Python statements (syntax) should be written using lowercase letters of the English.
7. IDLE?
   Integrated Development Environment (IDLE) is a text editor for running python instructions.
8. Variable? : A name used to store data temporarily in a program for later use.
   A variable is created using the assignment operator = . (eg:- age = 15).
9. **Operators** used in Python:
   Arithmetic (+, -, *, /, %), Assignment (=), Comparison (==, >) and Logical (and, or, not).
   Example:- The assignment operator = assigns a value to a variable, like A=20.
   The comparison operator == is used to compare two values to check if they are equal.
10. **input( )** function captures user input (as strings by default). **int()** for calculations.
    **print( )** displays the output.
    Example:- Name=input("Enter your name:") print(name, "Your name is:")

```
no_of_students = int(input("Enter number of students:"))
```

11. **if**, **elif** and **else** are used for conditional statements.
    Example:-
```
if age <= 14:
    print("You are in Junior Category")
else:
    print("You are in Sub-Junior Category")
```
12. **range( )** function generates a sequence of numbers.
    Exmple:-
```
for count in range(1, 11):
    print(count)
```
We will get the output 1,2,3,4,5,6,7,8,9,10
13. Loops (while loop and for loop) are used to repeat a set of instructions multiple times until a condition is met.
14. **list** option is possible to store different data in a single string.
    Example:- colors = [“red”,“yellow”, “blue”,“green”,“orange”,“violet”]
14. **turtle** module is used for drawing lines, shapes, and patterns in Python programming.
    For this, use: **from turtle import***
    Example: To ger a square having 100 length each side, use the following programme,
```
from turtle import*
forward(100)
right(90)
forward(100)
right(90)
forward(100)
right(90)
forward(100)
```
15. Some modules in Python:
    **turtle** : for graphics or drawing. **datetime** : for things related to date and time.
    **os** : related to operating system. **math** : for mathematical operation.



## Chapter 4 – Cyber Space

1. Cyberspace?
   A virtual space where computers, phones, servers, digital devices, networks and information are linked together.
   (this term was popularized by **William Gibson** in his novel titled *Neuromancer* in 1984)
2. Digital footprint?
   The set of information created while using the Internet.
3. Cyber identity?
   A person's digital representation in the cyber world.
4. Internet of Things (IoT)?
   A network of physical objects connected to the internet with sensors and software.
5. Gamification?
   Integration of game design elements in learning environments.
6. Cyber Infrastructure?
   Includes the internet, different hardware, network protocols, information, databases that store it, web pages, and security systems.
7. Metaverse?
   An immersive platform for gaming, social media, shopping, education using AR, VR, AI, and Blockchain.

8. Avatars in cyberspace?
Digital representations of users in virtual environments.
9. Unauthorized access to a computer or network?
Hacking.
10. Ethical hacking?
Using hacking techniques with owner's permission, to identify vulnerabilities.
11. Digital assets?
Financial assets that exist only in digital form.
Eg:- Digital currencies, digital wallets, points earned through online shopping, NFTs, domain names, and software.
12. Cryptocurrency?
A digital currency, operate on decentralized networks powered by advanced technology.
Eg:- Bitcoin, Ethereum, and Litecoin.
13. Transactions in cryptocurrency are secured using a technology called cryptography. (like Blockchain)
14. NFT (Non-Fungible Token)?
Refers the valuable things, which are proprietary in the cyber world.
Eg:- Digital art, digital versions of music, paintings, photographs, videos and in-game items.
15. Types of data collection?
Explicit and Implicit data collection.
16. Explicit data collection?
Occurs when a user knowingly provides their information. (through filling out forms, signing up for accounts, or giving consent for data collection).
17.Implicit data collection?
Information gathered without the user explicitly providing it. (through browsing history, location information, and details about devices connected to the internet)
18. Big Data?
Very large and complex datasets that traditional software cannot handle efficiently.
Eg:- Digital data generated by social media, e-commerce, and the IoT.
19. Plagiarism?
Presenting someone else's ideas or work as your own.
[Publishing someone else's study or work under your own name without citing the sources]
20. Problems caused due to the excessive use of internet?
FOMO (Fear of missing out), Gaming disorder,
Cyberchondria (madly searching for medical information online due to the fear of having symptoms of diseases),
Nomophobia (No Mobile Phobia: over anxiety when not having mobile phone),
Tech neck, Obesity and sedentary lifestyle.
21. Solution for unrestricted use of internet?
Limit screen time. Digital Detox (disconnect from the digital world every now and then).
Just keep a careful onlinepresence alone.
Follow healthy sleepinghabits.
Maintain strong personal relationships to balance online interactions.
22. Netiquette?
A set of rules for good behaviour on the Internet.
Eg:- Do not share harmful or hateful messages, comments, trolls, posts, news with others or other people's personal information.

## Chapter 5 – Make the Web Look Stylish

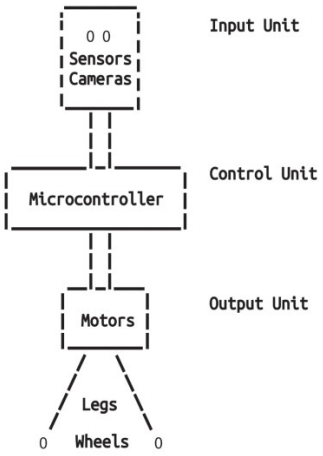
1. **HTML** (Hypertext Markup Language) is a markup language used to create web pages.
2. We need a **Text Editor** to include html tags for webpage making.
3. Some of the HTML tag have their closing tabes.
   Example:- Opening and closing tabs of Heading1 is <h1> and </h1>.
   Tags like <hr> for drawing a horizontal line and <br> for a line break are html elements that do not have closing tags or any content.
4. The ** ** tag is used to divide the code in a webpage into different sections (divisions).
5. <style> tag can be used to make the web page attractive.
6. Style instructions consist of two parts: the property of the style and its corresponding value.
   Example: In the tag, <h2 style="color: #0000FF;". Rasheed Odakkal <h2>,
   color: (the property of style) and #0000FF (the value)
7. <h2 style="**background-color:** #0000FF;".<h2> for getting the blue background.
8. To include image? Use <**img src=**"---">
   Example:- <img src="Gandhiji.png">
9. Difference between the inline cascading style and internal cascading style?
   **Inline cascading styles** involve adding styles within its own HTML tags.
   (Here, each tag needs to be styled everytime. Hence, codes are getting longer).
   **Internal cascading style** is the method of adding styles within the <style> tag .
   (Here, Style instructions are given inside the **{}** brackets within the <style> tag).
   Example:- CSS (cascading style sheets)
10. Advantage of CSS?
    CSS is convenient because this need to style a tag once (avoid repetition of style tags).
    In webpages, all the cascading styles we use are combined into a single file, given a name, and saved with a .css file extension. (When creating webpages, it is possible to include all these styles provided in this file using a single line of code)
11. **CSS** was introduced by the World Wide Web Consortium (W3C) in 1996. The idea was proposed by **Haakon Wiam Lee** and **Bert Bos**
12. Example for giving Internal style and External style?
    **<p style="background color:blue">** is an internal style.
    **<style> p { background color:blue }** is an external style.
13. The browser's priority for choosing the style is:
    External style → Internal style → Inline style.



## Chapter 6 – The World of Robots

1. **Robotics** ?
   The science that studies the design,construction, operation, and control of robots using appropriate software.
   Robotics combines knowledge from three diverse fields:
   Electronics, Mechanical Engineering and Computer Science
   The rapid advancement of Artificial Intelligence (AI) is driving significant progress in robotics.
2. Robots are machines that can perform complex tasks quickly and accurately for extended periods without tiring.
3. Robots recognize their surroundings using **sensors**.
   [Sense/Input → Analyze/Process → Decide/Output Action]

4. The basic components of a robotic system:




**Input Unit (Sensors)**: Gathers information from the surroundings (e.g., Light Sensor, IR Sensor, Microphone, Camera).

**Control Unit (Processor)**: Processes the gathered information, makes decisions based on instructions (e.g., Microcontrollers like Arduino, Raspberry Pi, ESP32).

**Output Unit (Actuators/Indicators)**: Performs actions or displays information based on the control unit's decisions (e.g., Motors, LEDs, Buzzers, Display Units)

5. The input devices and their uses:

**Light Sensor**: To detect the presence of light.

**IR Sensor**: To detect the presence of objects using infrared rays.

**Microphone**: To collect information in the form of sound.

**Camera**: To collect information in the form of images.

6. The **microcontrollers** used as control unit/the processor:

A microcontroller is a small computer integrated into a single chip. It contains the main components: **processor** (CPU), **RAM, ROM**, and **input/output interfaces**.

Examples: Arduino, Raspberry Pi, ESP32.

7. The output devices?

The **Actuators** or indicators, which are the components responsible for enabling automatic movement in robots.

Examples: Servo motors, Stepper motors.

8. Use of a servo motor?

To perform automatic or mechanical movements.

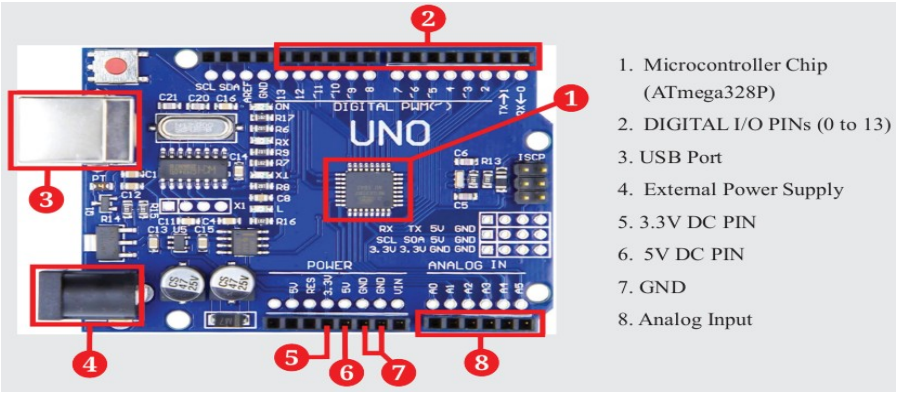
9. What is **Arduino**?

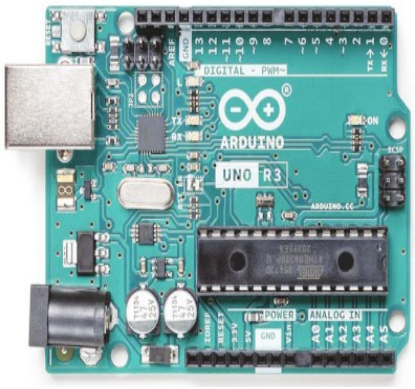
A world-famous open-source hardware/software platform.

Created to make physical computing devices that connect sensors and actuators easier, cheaper, and more popular to build. It's used to create prototypes or miniature versions of electronic devices.

10. The Arduino UNO board contains a microcontroller (ATmega328P).

11. The major components of Arduino UNO R3:



**Microcontroller Chip (ATmega328P)**: The brain; collects info and controls devices.

**DIGITAL I/O PINs (0 to 13)**: Collects data from input devices; controls output devices. Pins with ~ are for PWM.

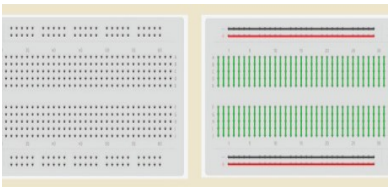
**USB Port** : Connects to computer to upload programs and transfer data.

**External Power Supply**: Supplies power from a battery or external source.

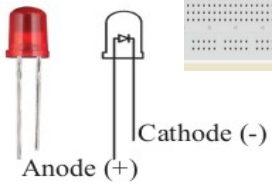
**5V PIN** / **3.3V PIN** : Provides a steady 5V or 3.3V, respectively.

**GND** : Provides the ground potential (**0V**).

12. Breadboard?
    A device that lets you build and reuse circuits without soldering. Electronic components can be attached to its holes.

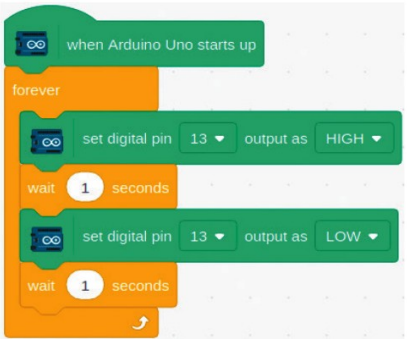


13. Connecting LED (Light Emiting Diode)?
    Connect the power's positive terminal to the LED's anode (+) and the power's negative terminal to the LED's cathode (-)




(Anode of LED is typically the longer terminal)

14. When connecting the LEDs,suitable **resistor** must be connected in series. Reason?
    To prevent excessive current flow and damage to the LED.
15. To blink the LED (turn it ON/OFF), it must be connected to?
    One of the Digital Pins (0 to 13).
16. To make an LED blink?
    Using the PictoBlox,
    - Set digital pin 13 output as HIGH turns the pin ON (LED lights up).
    - Set digital pin 13 output as LOW turns the pin OFF (LED goes out)




17. What is **PictoBlox**?
    A free, block-based visual programming platform, similar to Scratch, that teaches coding, AI, and robotics to beginners by letting them drag and drop colorful blocks to create games, animations, and control hardware like robots and IoT devices.
18. The two programming modes in PictoBlox?
    Upload Mode and Stage Mode.
    **1. Upload Mode**: The program is fully uploaded to the Arduino microcontroller's memory. A computer is NOT required to run the program later; only power is needed.




(this code only work in upload mode)

    **2. Stage Mode**: Arduino must first be configured using Upload Firmware. The Arduino runs according to instructions received via the USB cable. The program can only run while the Arduino is connected to the computer.




(this code only work in stage mode)

19. To make Beep sound at interval?
    - Connect the buzzer GND to Arduino GND.
    - Connect the I/O to the digital pin (eg:- 8) of the Arduino.
    - Connect VCC to the 5V of the Arduino.




**Buzzer**

20. Components needed for making an Automatic Sanitizer Dispenser:
    IR Sensor Module: To detect the presence of the hand (Input).
    Servo Motor: To control the tap (Actuator/Output).
    Arduino: To control the system logically (Control Unit).

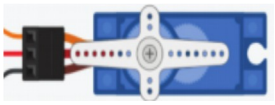


**IR Sensor**

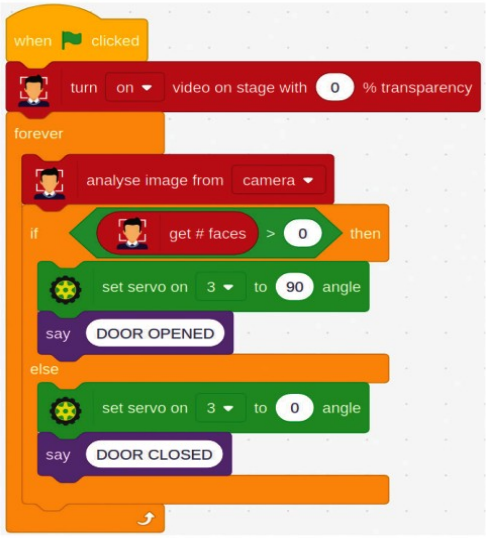
21. Principle behind the Automatic Sanitizer Dispenser:
    When an object (hand) is detected by the IR sensor, its output pin goes LOW (0). The Arduino reads this, and commands the servo motor to rotate (e.g., to 90 degrees) to open the tap. When the hand is removed, the IR sensor output goes HIGH (ON) , and the servo closes the tap (e.g., to 0 degrees)

22. Connection of different wires of a Servo Motor:
The red wire can be connected to the 5V of the Arduino.
The brown wire to the GND of the Arduino to provide power.
The orange wire is the control wire, determines the angle the output shaft should rotate.

**Servo Motor**



23. PIR sensor?
Passive infrared sensor. (can detects the infrared heat radiation emitted by the animals)
24. Technology behind Face-recognizing Smart door:
Computer Vision and Face Detection extension in PictoBlox.
25. For making a Face-recognizing Smart door, Stage Mode is selected in PictoBlox. Reason?
Stage Mode is used because the system relies on the laptop's camera.
(Arduino does not have a camera in it)




Mechanism:
The program uses the **get # faces** block
to check if a face is detected.
If the number of faces is greater than 0, the servo motor
rotates to 90 degrees (DOOR OPENED).
If no faces are detected, the servo motor
rotates to 0 degrees (DOOR CLOSED).



## Chapter 7 – Learning Science through Computers
## [Stellarium, QGIS, GeoGebra]

1. **Stellarium**?
A free desktop planetariumthat simulates views of the sky.
2. The major tools in the **Tool bar** of Stellarium?
Location Window, Date/Time Window, Search Window.
3. To Set the Location in Stellarium:
Click on the **Location Window** (F6) tool → Type location




4. To set the Observation day and time in Stellarium:
Click on the **Date/Time Window** (F5) tool → Set the year :Month: Day: Time.
(You can change the time by clicking on Increase Time Speed (L) in the status bar.
5. To search any celestial object like the Sun:
Click on the **Search Window** (F3) tool → Type the object name.
6. The following locations in the Stellarium software can be used as an observation point to view the midnight sun?
Tromso (in Norway) and Talnakh (in Russia).
7. **Quantum GIS**?
QGIS is a free Geographic Information System software.
**GRASS** open), **Arc GIS** (proprietary) are also Geographic Information System software.
8. To open QGIS? **QGIS → Project → Open**

9. To add a data in a new layer:
**Layer → Add Layer → Add Delimited Text Layer →Delimited Text →**brows the file.
Make the following settings in the window:

File Format : CSV (Comma Separated Value)
Geometry Definition : Point coordinates
X Field : Longitude, Y Field : Latitude
Geometry CRS : EPSG:4326 – WGS 84.

Click **Add**

10. To view information about any one point recorded on QGIS map:

Select the **new layer** →Click on the **Identify Features** tool and click on the point where you want to get the data.

11. To add a raster file to a QGIS map:
Click **Layer**→ **Add Layer → Add Raster Layer**, (make sure Source Type →File).
**Source → Raster Dataset**→ (Brows and add the file).

12. To print a map:
**Project →New Print Layout**
(You can use the **Add New Label** and **Add New Legend** for adding other information too)

13. **GeoGebra**?
A dynamic mathematics software suite used globally for teaching and learning STEM subjects.

14. Tool to mark a point? The **Point**  tool.
Tool to draw line? The **Line**  tool.

15. Method to draw a line without using the Line tool?
In the **Input Bar**, type the required index numbers.
Example 1: To draw a straight line through two points, A and B, in the Input bar, type:
A=(3,2) and press Enter,
B=(1,4) and press Enter,
f:x+y=5 and press Enter.
Example 2: To draw a segment (short line) through two points, A and B, type:
Segment(A,B) and press Enter.

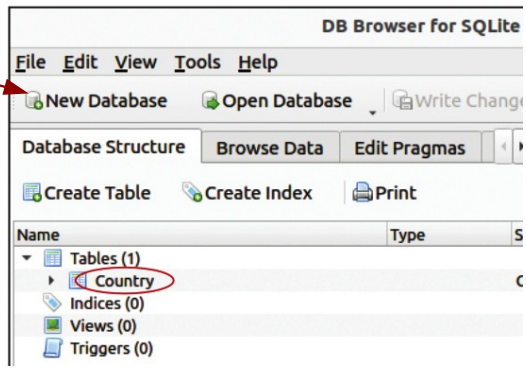
## Chapter 8 – Let's Give Life to Pictures

1. The basis of animation is **Persistence of vision** (remaining of image on the retina for one sixteenth of a second).
2. 2D animation softwares: Open Toonz, Tupi Tube Desk, Synfig Studio and Adobe Animate.
3D animation softwares: Blender, Autodesk Maya, and Adobe Dimension
3. Walt Disney's The Jungle Book, Bambi, Cinderella, etc. are famous 2D animated films.
Movies like Big Buck Bunny, Shrek, and Wall-E are famous 3D animated films.
4. **OpenToonz** is a free animation software based on "Toonz",developed by Digital Video S.p.A. in Italy.
5. The Japanese animation film 'Belle' (in 2021) was created using OpenToonz software.
6. To create a new animation project in Open Toonz, click **New Project**.
7. FPS (Frame per second)?
The number of frames displayed in one second.
8. **Xsheets** in OpenToonz?
Xsheets are a feature for organizing the content of animation scenes, similar to timelines in video editing software.

9. Examples for Rooms or Workspaces in Open Toonz?
   **Drawing** room and **Animation** room.
   Drawing room is used to draw, colour, and edit the drawings.
   Animation room is where you animate the position, rotation and size of the current objects.
10. Finding a story and detailed **storyboard** is important for the preparation of an animation.

## Chapter 9 – To the Information Storehouse

1. Oracle, MySQL, SQL Server, and sqlite are DBMS (Database Management system) applications.
   The information stored in DBMSs is presented to the user with the help of techniques like html,, Java, Python, etc.
2. **SQL** (Structured Query Language) is used to communicate with the DBMS.
   Operations like create and modify tables, insert information into tables, edit, delete information etc. can be done using SQL.
3. Sqlite (an open software) was developed by D.Richard Hipp.
   In SQlite, the database is stored as a single file on disk.
4. DB Browser for SQLite → **New Database** →
5. To Prepare Table: **Edit table (**give a name)→**Add**
6. Each column represents a Field.
   Each row of the table represents a Record.
   A table may contain several records.



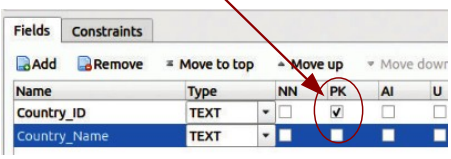

Player Table

| Player_ID | Player_Name | Total_Runs | Country_ID |
|---|---|---|---|
| P5001 | Sanath Jayasurya | 221 | C101 |
| P5002 | Sachin Tendulkar | 673 | C102 |
| P5003 | Lance Clusener | 281 | C105 |
| P5004 | Martin Crowe | 456 | C104 |
| P5005 | Glenn McGrath | 67 | C103 |

Fields

Record

7. For a record, if **Primary key** (PK) is set, it doesn't get repeated. (It will be distinguished)




8. Field Name  Datatype
   Name : TEXT
   Number : NUMERIC

9. To Add Records in DB Browser for SQLite?
   **Browse Data** tab →(select the table)→**Insert New Record**→Type data → **Write Changes**
10. Way to retrieve information or search data from the database?
    Use the **SQL Queries**.
    [Execute SQL tab → Type the SQL statement → select →Run].

## Chapter 10 – Software and Freedom

1. **Free Software** can be modified and redistributed (if its original license applies) by editing its source code.
2. Source code and object code of a programme?
   **Source code** is a human-readable instructions (written in a programming language).

**Object code** (or executable code) is a converted form of source code into machine executable format. (An **Interpreter** is used to convert a source code into an object code) Source code is editable, but object code cannot be easily modified.

3. "Code should be written in a readable and clear manner." Guide line of? Python programming.
4. **GPL** (General Public License) published by GNU can be used to redistribute a modified version.
   **Apache License**, which distributes the Android operating system, and the **BSD** (Berkeley Software Distribution) **License** are open source licenses.
5. **Unix** operating system (1969) built by Dennis Ritchie, Ken Thompson and Douglas McIlroy at AT&T.
6. **iOS** is the mobile OS of Apple Inc. (its kernal is Unix).
7. Mac OSX is by the Apple Inc Corporation. (its Kernal is XNU).
8. Proprietary softwares, Examples: Unix, Mac OSX, MS Windows...
   Free and open-source software (FOSS), Example: GNU/Linux.
8. **GNU Project** (1983) and **Free Software Foundation** (FSF , 1985) by **Richard Stallman**. (GNU = GNU is not Unix)
9. GNU/Linux Kernel is **Linux**, written by **Linus Torvalds** .
10. Copyleft is a licensing method that uses copyright law itself to ensure that all users retain fredom to use, copy, modify, and redistribute a program.



11. Two functional parts of OS:
    **Kernel**: -the part that directly interacts with the computer's hardware.
    **Shell** : -the part that interacts with the user. (GUI -graphical user interface is associated with the Shell.
12. Kernal of GNU/Linux OS is Linux,
    Shell (‘the Hurd’) of GNU/Linux OS is Bash (Bourne Again Shell)
13. GitHub, GitLab, GNU Savannah are popular online platforms that rely on the version control software for publishing and managing code.
14. The Linux versions Debian (from it Ubuntu, Kite GNU-Linux etc.), Red Hat linux and Slackware Linux are examples for free GNU/Linux OS distributions.
15. **Android** is a OS in touchscreen mobile phones, tablets etc. It is based on Linux kernel. (its program code is available under an open source license)
16. Arduino Uno board is an open hardware.

By **Rasheed Odakkal**
GVHSS Kondotty 9846626323

odakkal blog